

## ČECH COHOMOLOGY DAY 2

**DEFINITION:** Let  $M$  be an  $n$ -dimensional manifold. An open cover  $\mathfrak{U} := \{U_\alpha\}_{\alpha \in I}$  of  $M$  is called a *good cover* if all finite intersections  $U_{\alpha_1} \cap \cdots \cap U_{\alpha_t}$  are diffeomorphic to  $\mathbb{R}^n$ .

**RECALL [ČECH COHOMOLOGY]:** Let  $\mathcal{F}$  be a presheaf on a topological space  $X$  and  $\mathfrak{U} := \{U_\alpha\}_{\alpha \in I}$  be a [countable and ordered] open cover of  $X$ . We put

$$C^p(\mathfrak{U}, \mathcal{F}) = \prod_{\alpha_0 < \alpha_1 < \cdots < \alpha_p} \mathcal{F}(U_{\alpha_0 \alpha_1 \cdots \alpha_p})$$

where  $U_{\alpha_0 \alpha_1 \cdots \alpha_p} = U_{\alpha_0} \cap U_{\alpha_1} \cap \cdots \cap U_{\alpha_p}$ . Each inclusion  $\partial_i : U_{\alpha_0 \cdots \alpha_p} \hookrightarrow U_{\alpha_0 \cdots \hat{\alpha}_i \cdots \alpha_p}$  induces a restriction map  $\mathcal{F}(\partial_i)$ , and we make  $C^*(\mathfrak{U}, \mathcal{F})$  a complex by defining  $\delta^p : C^p(\mathfrak{U}, \mathcal{F}) \rightarrow C^{p+1}(\mathfrak{U}, \mathcal{F})$  by  $\delta^p = \mathcal{F}(\partial_0) - \mathcal{F}(\partial_1) + \cdots + (-1)^{p+1} \mathcal{F}(\partial_p)$ .

The cohomology of the above complex, denoted  $H^*(\mathfrak{U}, \mathcal{F})$ , is the *Čech cohomology of the cover  $\mathfrak{U}$  with values in  $\mathcal{F}$* .

Consider  $S^1$  with the usual topology inherited from  $\mathbb{R}^2$  and endow  $\mathbb{Z}$  the discrete topology. Let  $\mathcal{F}$  be the sheaf on  $S^1$  that assigns each open set  $U$  to  $\mathbb{Z}$ .

**EXERCISE 1.** Recall (or recompute) the Čech cohomology of  $S^1$  with the sheaf as described above with respect to the cover  $U_1 = S^1 \setminus \{(0, 1)\}$  and  $U_2 = S^1 \setminus \{(0, -1)\}$ .

**EXERCISE 2.** Compute the Čech cohomology of  $S^1$  with the sheaf as described above with respect to the cover [with one element]  $U = \{\vec{x} \in \mathbb{R}^2 : 1/2 < \|\vec{x}\| < 3/2\}$ .

**THEOREM:** If  $\mathfrak{U}$  is a good cover of a manifold  $M$ , then the de Rham cohomology of  $M$  is isomorphic to the Čech cohomology of the good cover:

$$H_{\text{DR}}(M) \cong H(\mathfrak{U}, \mathbb{R}).$$

**REMARK:** This Theorem is given prior to the discussion of (pre)sheaves. Here,  $H(\mathfrak{U}, \mathbb{R})$  denotes the homology of the Čech complex  $C^*(\mathfrak{U}, \mathbb{R})$ , where the  $p$ th cochain space  $C^p(\mathfrak{U}, \mathbb{R})$  consists of the locally constant functions on the  $(p+1)$ -fold intersections of the members of the cover  $\mathfrak{U}$ .

**COROLLARY:** The Čech cohomology  $H(\mathfrak{U}, \mathbb{R})$  is the same for all good covers  $\mathfrak{U}$  of  $M$ .

An outline for the proof of the THEOREM:

- We wish to generalize the usual Mayer-Vietoris sequence (with two open sets) to a setting where one has countably many open sets rather than just two. Given a countable cover  $\mathfrak{U} = \{U_\alpha\}_{\alpha \in I}$ , the inclusions  $U_{\alpha_0 \cdots \alpha_t} \subseteq U_{\alpha_0 \cdots \hat{\alpha}_i \cdots \alpha_t}$  induce restrictions of forms

$$\prod_{\alpha_1 < \alpha_2 < \cdots < \alpha_t} \Omega^*(U_{\alpha_1 \cdots \hat{\alpha}_i \cdots \alpha_n}) \xrightarrow{\delta} \prod_{\alpha_1 < \alpha_2 < \cdots < \alpha_t} \Omega^*(U_{\alpha_1 \cdots \alpha_t}).$$

Here, as before,  $\Omega(U) = C^\infty(U) \otimes \Omega^*$  is the usual object of study from De Rham cohomology.

- It is the content of Prop 8.5 (Bott-Tu) that the constructed generalized Mayer-Vietoris sequence

$$0 \rightarrow \Omega^*(M) \rightarrow \prod_{\alpha \in I} \Omega^*(U_\alpha) \rightarrow \prod_{\alpha < \beta} \Omega^*(U_{\alpha\beta}) \rightarrow \prod_{\alpha < \beta < \gamma} \Omega^*(U_{\alpha\beta\gamma}) \rightarrow \cdots$$

is indeed an exact sequence [for any countable open cover  $\{U_\alpha\}_{\alpha \in I}$  of a manifold  $M$ ]. For those curious, the argument involves partitions of unity.

- One can now furnish a double complex  $C^*(\mathfrak{U}, \Omega^*)$  as shown below.

$$\begin{array}{ccccccc}
& \vdots & & \vdots & & \vdots & \\
& \uparrow & & \uparrow & & \uparrow & \\
0 & \longrightarrow & \prod_{\alpha \in I} \Omega^2(U_\alpha) & \xrightarrow{\delta} & \prod_{\alpha < \beta} \Omega^2(U_{\alpha\beta}) & \xrightarrow{\delta} & \prod_{\alpha < \beta < \gamma} \Omega^2(U_{\alpha\beta\gamma}) \longrightarrow \cdots \\
& \uparrow d & & \uparrow d & & \uparrow d & \\
0 & \longrightarrow & \prod_{\alpha \in I} \Omega^1(U_\alpha) & \xrightarrow{\delta} & \prod_{\alpha < \beta} \Omega^1(U_{\alpha\beta}) & \xrightarrow{\delta} & \prod_{\alpha < \beta < \gamma} \Omega^1(U_{\alpha\beta\gamma}) \longrightarrow \cdots \\
& \uparrow d & & \uparrow d & & \uparrow d & \\
0 & \longrightarrow & \prod_{\alpha \in I} \Omega^0(U_\alpha) & \xrightarrow{\delta} & \prod_{\alpha < \beta} \Omega^0(U_{\alpha\beta}) & \xrightarrow{\delta} & \prod_{\alpha < \beta < \gamma} \Omega^0(U_{\alpha\beta\gamma}) \longrightarrow \cdots \\
& \uparrow & & \uparrow & & \uparrow & \\
& 0 & & 0 & & 0 & 
\end{array}$$

The  $\delta$  maps are as we defined before and the  $d$  maps are the usual differentials in the de Rham complex. Of course, there are some signs that should be introduced so that this is a true double complex [because we hope to eventually take the totalization]. By abuse of notation, the totalization of the double complex  $\text{Tot}^\oplus(C^*(\mathfrak{U}, \Omega^*))$  [where the  $n$ th cochain space of  $\text{Tot}^\oplus(C^*(\mathfrak{U}, \Omega^*))$  is  $\bigoplus_{n=p+q} C^p(\mathfrak{U}, \Omega^q)$ ] is also denoted at times as  $C^*(\mathfrak{U}, \Omega^*)$ .

- Further, one can augment the columns by the Čech complex  $C^*(\mathfrak{U}, \mathbb{R})$  and the rows by the De Rham complex  $\Omega^*(M)$ . One can check that since the columns are exact, the cohomology of the totalized complex is isomorphic to the homology of the first column, i.e. the De Rham cohomology.
- Moreover, when the cover  $\mathfrak{U}$  is “nice enough” [e.g. a good cover] then the columns of the double complex are exact too. Whence, one argues that the cohomology of the totalization also computes the cohomology of the first row, i.e. the Čech cohomology. It follows that Čech and De Rham cohomology are isomorphic.

**DEFINITION:** Let  $\mathfrak{U} = \{U_\alpha\}_{\alpha \in I}$  be an open cover of a manifold  $M$ . An open cover  $\mathfrak{V} = \{V_\beta\}_{\beta \in J}$  of  $M$  is called a *refinement* of  $\mathfrak{U}$ , which we may write as  $\mathfrak{U} < \mathfrak{V}$ , if there is a map  $\phi : J \rightarrow I$  such that  $V_\beta \subseteq U_{\phi(\beta)}$ . Note  $\phi$  need not be injective or surjective.

**REMARK:** In the setup above, the map  $\phi$  making  $\mathfrak{V}$  a refinement of  $\mathfrak{U}$  induces a map

$$\begin{aligned}
\phi^\# : C^p(\mathfrak{U}, \mathcal{F}) &\rightarrow C^p(\mathfrak{V}, \mathcal{F}) \\
(\phi^\# \omega)(V_{\beta_1 \dots \beta_p}) &= \omega(U_{\phi(\beta_0) \dots \phi(\beta_p)}).
\end{aligned}$$

**LEMMA:**  $\phi^\#$  is a chain map.

**LEMMA:** Given an open cover  $\mathfrak{U} = \{U_\alpha\}_{\alpha \in I}$  and a refinement  $\mathfrak{V} = \{V_\beta\}_{\beta \in J}$ , if  $\phi, \psi : J \rightarrow I$  are two refinement maps, then  $\phi^\#$  and  $\psi^\#$  are homotopic.

**EXERCISE 3.** Prove that  $\phi^\#$  is a chain map [i.e. that  $\phi^\#$  commutes with the differential  $\delta$  on  $C^*(\mathfrak{U}, \mathcal{F})$ ].

**EXERCISE 4.** With the setup of the second lemma, put  $K : C^q(\mathfrak{U}, \mathcal{F}) \rightarrow C^{q-1}(\mathfrak{V}, \mathcal{F})$  defined by  $(K\omega)(V_{\beta_1 \dots \beta_q}) = \sum_i (-1)^i \omega(U_{\phi(\beta_0) \dots \phi(\beta_i) \psi(\beta_i) \dots \psi(\beta_q)})$ . Show that  $\psi^\# - \phi^\# = \delta K + K\delta$ , and therefore  $\phi^\#$  and  $\psi^\#$  are homotopic.

DEFINITION. By the two lemmas above, given a refinement  $\mathfrak{U} < \mathfrak{V}$  of open covers on a manifold  $X$ , there is a well-defined map on cohomology  $H^*(\mathfrak{U}, \mathcal{F}) \rightarrow H^*(\mathfrak{V}, \mathcal{F})$  that makes  $\{H^*(\mathfrak{U}, \mathcal{F})\}_{\mathfrak{U} \in \mathcal{U}}$  a directed system of cohomology groups. The direct limit of this system

$$\check{H}^*(X, \mathcal{F}) := \varinjlim_{\mathfrak{U}} H^*(\mathfrak{U}, \mathcal{F})$$

is called the *Čech cohomology of  $X$  with values in the presheaf  $\mathcal{F}$* .

PROPOSITION. Let  $\mathbb{R}$  be the constant presheaf on a manifold  $M$ . Then the Čech cohomology of  $M$  with values in  $\mathbb{R}$  is isomorphic to the de Rham cohomology

$$\check{H}^*(M, \mathbb{R}) \cong H_{\text{DR}}^*(M).$$

EXERCISE 5. Check that the systems of groups and homomorphisms used in the above definition of Čech cohomology with values in a specified presheaf  $\mathcal{F}$  is indeed a direct system.